Swiggy Data Analysis Using SQL (End-to-End Data Analytics Project)

Introduction: This project analyses Swiggy food delivery data using SQL to extract meaningful insights such as top-performing restaurants, most ordered dishes, and revenue trends.

Data Cleaning and Validation:

- **null check : if there are any null it returns 1 , if not there then returns 0**

```
select
    sum(case when State IS NULL THEN 1 ELSE 0 END) AS null_state,
    sum(case when City IS NULL THEN 1 ELSE 0 END) AS null_city,
    sum(case when Order_Date IS NULL THEN 1 ELSE 0 END) AS null_order_date,
    sum(case when Restaurant_Name IS NULL THEN 1 ELSE 0 END) AS null_restaurant,
    sum(case when Location IS NULL THEN 1 ELSE 0 END) AS null_location,
    sum(case when Category IS NULL THEN 1 ELSE 0 END) AS null_category,
    sum(case when Dish_Name IS NULL THEN 1 ELSE 0 END) AS null_dish,
    sum(case when Price_INR IS NULL THEN 1 ELSE 0 END) AS null_price,
    sum(case when Rating IS NULL THEN 1 ELSE 0 END) AS null_rating,
    sum(case when Rating_Count IS NULL THEN 1 ELSE 0 END) AS null_rating_count
from swiggy_data
```

- **OutPut:**

Results | Messages

	null_state	null_city	null_order_date	null_restaurant	null_location	null_category	null_dhis	null_price	null_rating	null_rating_count
1	0	0	0	0	0	0	0	0	0	0

Insight:

- All columns have 0 null values
- Dataset is clean and ready for analysis

- **checking blank string or empty strings**

```
select * from swiggy_data
```

where State = '' OR city = '' OR Restaurant_Name = '' OR Category = '' OR Dish_Name = '';

- **OutPut:**

Results | Messages

	State	City	Order_Date	Restaurant_Name	Location	Category	Dish_Name	Price_INR	Rating	Rating_Count

- **Insight:**
  • The dataset is complete with no missing or empty values, making it suitable for analysis.

➢ **CHECK for duplicates values**

```
select

state,city,order_date,Restaurant_name,location,category,

dish_name,price_INR,rating,rating_count,count(*) as CNT

FROM swiggy_data

group by

state,city,order_date,Restaurant_name,location,category,

dish_name,price_INR,rating,rating_count

having count(*) > 1
```

- **OutPut:**

Results | Messages

	State	City	Order_Date	Restaurant_Name	Location	Category	Dish_Name	Price_INR	Rating	Rating_Count

- **Insight:**
  • No duplicate records were found, ensuring data integrity.

➢ **Deletion of duplicate values**

```
with CTE AS (

select * ,

ROW_NUMBER() over(

 partition by state,city,order_date,Restaurant_name,location,category,

dish_name,price_INR,rating,rating_count

order by (select null)
```

```
) as rn

from swiggy_data

)

DELETE FROM CTE WHERE rn>1
```

- **OutPut:**

```
Messages

(0 rows affected)

Completion time: 2026-03-22T18:41:10.4360114+05:30
```

- **Insight:**
  • No duplicate records
  • Data uniqueness maintained

Data Modelling (Star Schema)

- The dataset was transformed into a star schema model to improve query performance and simplify analysis.
- A central fact table (fact_swiggy_orders) was created to store transactional data such as price, rating, and rating count.
- Dimension tables were created for Date, Location, Restaurant, Category, and Dish to organize descriptive attributes.

Fact Table:

fact_swiggy_orders

Dimension Tables:
• dim_date
• dim_location
• dim_restaurant
• dim_category
• dim_dish

Requirements

➢ **Key Performance Indicators (KPIs)**

- **TOTAL ORDERS**

```
SELECT COUNT(*) AS totalorders

from fact_swiggy_orders
```

- **OutPut:**

Results | Messages

	totalorders
1	197401

- **TOTAL REVENUE**

```
select

format(sum(convert(float,price_INR))/1000000,'N2')  +  ' INR_MILLION'  as total_revenue

from fact_swiggy_orders
```

- **OutPut:**

Results | Messages

	total_revenue
1	53.00 INR_MILLION

- **AVG DISH PRICE**

```
select

format(avg(convert(float,price_INR)),'N2')  +  ' INR'  as avg_revenue

from fact_swiggy_orders
```

- **OutPut:**

Results | Messages

	avg_dishPrice
1	268.50 INR

- **AVG RATING**

```
select

avg(rating) as avg_rating

from fact_swiggy_orders
```

- **OutPut:**

Results | Messages

	avg_rating
1	4.341577

Business Analysis (Deep Dive)

- **MONTHLY ANALYSIS**

```
select
d.year,
d.month,
d.month_name,
sum(price_INR) as total_revenue
from fact_swiggy_orders f
JOIN dim_date d
on f.Date_id = d.date_id
group by
d.year,
d.month,
d.month_name
order by total_revenue desc
```

- **OutPut:**

Results | Messages

	year	month	month_name	total_revenue
1	2025	1	January	6823981.33
2	2025	5	May	6792621.35
3	2025	8	August	6790899.22
4	2025	7	July	6650268.51
5	2025	4	April	6590449.00
6	2025	3	March	6572738.20
7	2025	6	June	6513676.19
8	2025	2	February	6268106.67

- **QUATERLY ANALYSIS**

```
select
d.year,
d.Quarter,
sum(price_INR) as total_revenue
from fact_swiggy_orders f
JOIN dim_date d
on f.Date_id = d.date_id
group by
d.year,
d.Quarter
order by total_revenue desc
```

- **OutPut:**

Results | Messages

	year	Quarter	total_revenue
1	2025	2	19896746.54
2	2025	1	19664826.20
3	2025	3	13441167.73

- **YEARLY TREND**

```
select
d.year,
count(*) as totalorders
from fact_swiggy_orders f
JOIN dim_date d
on f.Date_id = d.date_id
group by
d.year
order by totalorders desc
```

- **OutPut:**

Results | Messages

	year	totalorders
1	2025	197401

- **ORDERS BY DAYS AND WEEKS**

```
select
DATENAME(WEEKDAY,d.Full_date) as day_name,
count(*) as totalorders
from fact_swiggy_orders f
JOIN dim_date d
on f.Date_id = d.date_id
group by
DATENAME(WEEKDAY,d.Full_date), Datepart(WEEKDAY,d.Full_Date)
order by DATEPART(WEEKDAY,d.Full_date)
```

- **OutPut:**

Results | Messages

	day_name	totalorders
1	Sunday	28469
2	Monday	27568
3	Tuesday	27413
4	Wednesday	28284
5	Thursday	28450
6	Friday	28284
7	Saturday	28933

LOCATION BASED ANALYSIS

- **TOP 10 CITIES BY ORDERS**

```
select top 10
l.city ,
sum(f.price_INR)  as total_revenue
from fact_swiggy_orders f
JOIN dim_location l
on f.location_id = l.location_id
group by l.City
order by sum(f.price_INR) desc
```

- **OutPut:**

Results | Messages

	city	total_revenue
1	Bengaluru	5455887.73
2	Lucknow	3117359.65
3	Hyderabad	3021656.62
4	Mumbai	3015573.35
5	New Delhi	2829180.60

- **REVENUE CONTRIBUTION BY STATE WISE**

```
select top 5
l.State ,
sum(f.price_INR)  as total_revenue
from fact_swiggy_orders f
JOIN dim_location l
on f.location_id = l.location_id
group by l.State
order by sum(f.price_INR) desc
```

- **OutPut:**

Results | Messages

	State	total_revenue
1	Karnataka	5455887.73
2	Uttar Pradesh	3117359.65
3	Telangana	3021656.62
4	Maharashtra	3015573.35
5	Delhi	2829180.60

- **TOP 5 RESTAURANTS BY ORDERS**

```
select top 10
r.restaurant_name ,
sum(f.price_INR)  as total_revenue
from fact_swiggy_orders f
JOIN dim_restaurant r
on f.restaurant_id = r.restaurant_id
group by r.Restaurant_Name
order by sum(f.price_INR) desc
```

- **OutPut:**

Results | Messages

	restaurant_name	total_revenue
1	KFC	4245461.78
2	McDonald's	3342455.58
3	Pizza Hut	2133265.69
4	Burger King	1900518.09
5	Domino's Pizza	1832985.32

- **TOP 5 CATEGORIES BY ORDER VOLUM**

```
select top 5
c.category ,
sum(f.price_INR)  as total_revenue
from fact_swiggy_orders f
JOIN dim_category c
on f.category_id = c.category_id
group by c.category
order by sum(f.price_INR) desc
```

- **OutPut:**

Results | Messages

	category	total_revenue
1	Recommended	7188217.53
2	Main Course	767175.00
3	BURGERS	694960.48
4	Burger Combos (3 Pc Meals)	507773.55
5	Desserts	500588.72

- **MOST ORDERED DISH**

```
select top 5
dd.dish_name,
sum(f.price_INR)  as total_revenue
from fact_swiggy_orders f
JOIN dim_dish dd
on f.dish_id = dd.dish_id
group by dd.Dish_Name
order by sum(f.price_INR) desc
```

- **OutPut:**

Results | Messages

	dish_name	total_revenue
1	Bold BBQ Veggie Thin n Crispy	99617.00
2	Korean & Thai Roll Chicken Meal	95591.57
3	Full House Popcorn Chicken Bucket	79200.00
4	Paneer Butter Masala	76181.82
5	Triple Chicken Feast	70757.00

- **Cuisine Performance (order + avg rating)**

```
select top 10
c.category,
count(*)  as total_orders,
avg(f.rating) as avg_rating
from fact_swiggy_orders f
JOIN dim_category c
on f.category_id = c.category_id
group by c.Category
order by count(*) desc
```

- **OutPut:**

Results | Messages

	category	total_orders	avg_rating
1	Recommended	24097	4.321782
2	Desserts	3582	4.371635
3	Main Course	2983	4.310191
4	Beverages	2682	4.368717
5	BURGERS	2538	4.324940

- **Total Orders by Range**

```
SELECT
   CASE
      WHEN CONVERT(FLOAT, Price_INR) < 100 THEN 'Under 100'
      WHEN CONVERT(FLOAT, Price_INR) BETWEEN 100 AND 199 THEN '100 - 199'
      WHEN CONVERT(FLOAT, Price_INR) BETWEEN 200 AND 299 THEN '200 - 299'
      WHEN CONVERT(FLOAT, Price_INR) BETWEEN 300 AND 499 THEN '300 - 499'
      ELSE '500+'
   END AS price_range,
   COUNT(*) AS total_orders
FROM fact_swiggy_orders
GROUP BY
   CASE
      WHEN CONVERT(FLOAT, Price_INR) < 100 THEN 'Under 100'
      WHEN CONVERT(FLOAT, Price_INR) BETWEEN 100 AND 199 THEN '100 - 199'
```

```
        WHEN CONVERT(FLOAT, Price_INR) BETWEEN 200 AND 299 THEN '200 - 299'
        WHEN CONVERT(FLOAT, Price_INR) BETWEEN 300 AND 499 THEN '300 - 499'
        ELSE '500+'
    END
ORDER BY total_orders DESC;
```

- **OutPut:**

Results | Messages

	price_range	total_orders
1	100 - 199	56189
2	200 - 299	54567
3	300 - 499	43758
4	Under 100	26795
5	500+	16092

- **Rating Analysis**

```
select
rating,
count(*) as totalorders
from fact_swiggy_orders
group by rating
order by rating desc
```

- **OutPut:**

Results | Messages

	rating	totalorders
1	5.00	9401
2	4.90	5713
3	4.80	8809
4	4.70	9089
5	4.60	10840
6	4.50	9946
7	4.40	85642
8	4.30	13698
9	4.20	8214
10	4.10	7619
11	4.00	5346
12	3.90	4021
13	3.80	3966
14	3.70	2711
15	3.60	2010
16	3.50	1883
17	3.40	1340

CONCLUSION SECTION

This project demonstrates how SQL can be used to transform raw data into meaningful business insights.

By performing data cleaning, modelling, and analysis, we identified key trends in customer behaviour, top-performing restaurants, and revenue distribution. These insights can help businesses make data-driven decisions.